

Program simulating semi-analytically the magnetotelluric response of a sphere in a layered earth

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Running head: **Magnetotelluric sphere program**

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ABSTRACT

A new program has been developed to simulate the magnetotelluric (MT) response of a sphere located in a layered earth. Originally designed for airborne electromagnetic (EM) exploration, where the source is a dipole, it has been adapted to plane waves and coded in Fortran and Python. Based on developments in the evaluation of derivatives with expansions of spherical and cylindrical functions, this program is open source and can be used to validate numerical models or to better understand the MT response of buried conductors in a layered environment.

INTRODUCTION

Various three-dimensional models have been developed to interpret magnetotelluric data (Ting and Hohmann, 1981; Wannamaker et al., 1984; Mackie et al., 1993; Weiss and Newman, 2002; Farquharson and Miensopust, 2011; Chave and Jones, 2012; Bello et al., 2019; Ansari et al., 2020, and others). As there are few analytical solutions available for these three dimensional models, the validation procedure is usually to compare solutions obtained with different algorithm (Mackie et al., 1993; Siripunvaraporn et al., 2002; Bello et al., 2019). We propose as a tool for this task the adaptation of a recently developed semi-analytic model of a sphere in a layered earth, developed for airborne applications (Vallée, 2015; Vallée and Moussaoui, 2023).

While there are several analytic solutions to the response of a conducting sphere to a magnetic dipole (March, 1953; Wait, 1953; Grant and West, 1965; Lodha and West, 1976; Ward and Hohmann, 1987), the sphere must be located in a uniform whole space, which is either resistive or conductive (Singh, 1973). A new approach has been initiated by Gumerov and Duraiswami (2004, 2007), who developed new relations describing derivatives of spherical functions as function expansions. Their algorithm leads to a new approach to EM analytic modelling where the goal is not to derive EM fields from potentials by symbolic manipulation, but to represent the fields as sums of analytic functions whose coefficients are related to coefficient potentials. Vallée and Moussaoui (2023) demonstrated that this approach can be extended to cylindrical functions, from which they develop a relationship between cylindrical and spherical functions. This allows the representation of the spherical fields originating from a

sphere target as cylindrical fields which are evaluated with already developed Hankel transform digital filters (Christensen, 1990; Werthmüller, 2017). This material has been successfully used in an algorithm for modelling the AEM response of a sphere in a layered earth under a finite source.

This model is interesting for airborne EM survey interpretation as the source and receiver are at some distance from the target, but less interesting for magnetotelluric routine interpretation where receivers are located near topographic variations and shallow conductors, with the exception being the use of the sphere model to study galvanic distortion (Groom and Bailey, 1991). However, it has applications for calibration of numerical models which consider the conductivity contrast between a localized body and its environment. It can also be used for heuristic studies of such models. Using the development of plane-wave fields in a layered earth and their spherical representations developed by Ward and Hohmann (1987), it is possible to adapt the algorithm developed by Vallée and Moussaoui (2023) to magnetotelluric modelling. On these foundations, we developed an open source program, available in Fortran and Python, for modelling the magnetotelluric response of a sphere in a layered earth. The major difference in the approach of this paper compared to Vallée and Moussaoui (2023) is the prediction of the primary field, in occurrence a plane wave field in a layered earth, and its representation as spherical potentials, as well as the computation of the magnetotelluric fields at surface. After presenting the theory, supported by numerous appendices, we describe the algorithm supporting the program, followed by synthetic examples showing some applications, and a comparison

of results of the semi-analytic solution with finite element (FE) numerical modelling.

THEORY

The initial impetus of this software has been the work of Gumerov and Duraiswami (2004, 2007). They demonstrated that derivatives of spherical functions, which are used to develop analytic solutions for electromagnetic fields, can be represented as expansions of spherical functions such as, (representing horizontal derivatives as $\frac{\partial}{\partial x} \pm i \frac{\partial}{\partial y}$),

$$\left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) T_n^m(\mathbf{r}) = k \left[b_{n+1}^{-m-1} T_{n+1}^{m+1}(\mathbf{r}) - b_n^m T_{n-1}^{m+1}(\mathbf{r}) \right] \quad (1)$$

$$\left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) T_n^m(\mathbf{r}) = k \left[b_{n+1}^{m-1} T_{n+1}^{m-1}(\mathbf{r}) - b_n^{-m} T_{n-1}^{m-1}(\mathbf{r}) \right] \quad (2)$$

$$\frac{\partial}{\partial z} T_n^m(\mathbf{r}) = k \left[a_{n-1}^m T_{n-1}^m(\mathbf{r}) - a_n^m T_{n+1}^m(\mathbf{r}) \right] \quad (3)$$

where $T_n^m(\mathbf{r})$ is either a regular or a singular spherical function at location $\mathbf{r}$, with n and m representing the degree and order, respectively (defined in Appendix B), $k = \sqrt{\mu\epsilon\omega^2 - i\mu\sigma\mu}$ is the EM propagation constant, μ the magnetic permeability, ϵ the electric permittivity, ω the radial frequency, and a_n^m and b_n^m are real coefficients which can be evaluated from the following expressions as

$$a_n^m = \begin{cases} \sqrt{\frac{(n+1+|m|)(n+1-|m|)}{(2n+1)(2n+3)}}, & |m| \leq n \\ 0, & |m| > n \end{cases} \quad (4)$$

$$b_n^m = \begin{cases} \sqrt{\frac{(n-m-1)(n-m)}{(2n-1)(2n+1)}}, & 0 \leq m \leq n \\ -\sqrt{\frac{(n-m-1)(n-m)}{(2n-1)(2n+1)}}, & -n \leq m < 0 \\ 0, & |m| > n \end{cases} \quad (5)$$

These relations can be applied to various differential operators and in particular used to develop field expansions representing the EM field Cartesian components. For example, for the vertical components, which are used in modelling the EM response of a sphere in a layered earth (Vallée and Moussaoui, 2023)

$$(E_z)_n^m = im(\psi_f)_n^m + \frac{k}{\hat{y}} \left[(n+2)a_n^m(\psi_a)_{n+1}^m + (n-1)a_{n-1}^m(\psi_a)_{n-1}^m \right] \quad (6)$$

$$(H_z)_n^m = -im(\psi_a)_n^m + \frac{k}{\hat{z}} \left[(n+2)a_n^m(\psi_f)_{n+1}^m + (n-1)a_{n-1}^m(\psi_f)_{n-1}^m \right] \quad (7)$$

where $(\psi_a)_n^m$ are $(\psi_f)_n^m$ the TM and TE electric potentials (defined in Appendix B), $\hat{y} = \sigma + i\epsilon\omega$ is the admittivity, and $\hat{z} = i\omega\mu$ is the impedivity. This provides a new approach to analytical or quasi-analytical EM modelling where fields can be related directly to potentials, instead of processing through a symbolic operation. Furthermore, Vallée and Moussaoui (2023) showed that this approach could be extended to cylindrical functions and were able to link spherical and cylindrical functions (described in Appendix D), opening avenues for computing semi-analytical solutions of the response of a sphere in a layered earth.

The application of this approach to magnetotelluric modelling requires using various algorithms that are dispersed throughout the literature and are presented in the Appendices for reference. First, we need to describe mathematically all the EM plane wave fields propagating in a layered earth (Appendix A). Representation of these fields by spherical expansions is also required in the modelling of the magnetotelluric response of a sphere (Appendix B). Then, the interaction of these fields with the anomalous sphere is captured by the reflection coefficients described in Appendix C. Appendix D demonstrates the possibility to represent derivatives of cylindrical waves

as function expansions and their relations to spherical functions. This has an impact in various ways. Firstly, it allows the representation of spherical fields as cylindrical expansions. Secondly, it allows the use of a description of the propagation of the EM fields in the layered earth as TM and TE potentials as proposed by Ward and Hohmann (1987), because the fields can be computed as function expansions directly from potentials functions expansions in wavenumber domain where the computation of the layered-earth coefficients takes place (Appendix E). Finally, Vallée and Mousaoui (2023) discuss the impact that the layered response has on the sphere field and propose an algorithm to correct this effect (Appendix F).

ALGORITHM

The program that we developed is built around these algorithms, and the flow chart shown in Figure 1 is linked to the following discussion as item numbers refer to the steps of the flow chart.

1. Initial parameters read by the program include measurement locations, frequencies, layered earth (thicknesses and conductivities) and sphere (radius, conductivity, depth) parameters.
2. For each frequency, computation of the sphere reflection coefficients (Ward and Hohmann, 1987, p. 301), as this computation is independent of the measurement locations. Reflection coefficients relate inward and outward spherical fields at the boundary of a sphere, while satisfying the continuity criteria of the tan-

gential electric and magnetic fields. They have been derived by various authors including March (1953); Wait (1953); Ward and Hohmann (1987) and are presented in Appendix C for spherical Bessel functions.

3. For each frequency, computation of the layered earth reflection correction developed by Vallée and Moussaoui (2023), is done once as this computation is independent of the measurement locations. This layered-earth correction compensates for the field originating from the sphere which is reflected by the layer boundaries. The mathematical details are presented in Appendix F. The computation of this correction requires a spherical representation of the fields emanated from the layered earth on the sphere, which is accomplished using spherical harmonic analysis (Adams and Swarztrauber, 1999).
4. Computation of the primary fields on the surface of the sphere following plane-wave layered earth propagation equations (Ward and Hohmann, 1987, p. 194-199). In magnetotelluric, the incident field is a plane wave. For the purpose of modelling the response of an anomalous object, the first step is to model the primary field in the subsurface. Propagation of a plane wave in a layered earth is a known topic and we follow in Appendix A the development of Ward and Hohmann (1987) while adding required knowledge on the upward and downward primary fields in each layer.
5. Representation of the primary fields on the surface of the sphere as spherical coefficients of radial electric and magnetic field potentials. Ward and Hohmann (1987), pp. 292-296, represent a plane wave in a uniform media as a sum of

spherical harmonics. Their results are reproduced in Appendix B and extended to upward propagation and orthogonal electric field. Furthermore, as the primary field in a layered earth is represented as upward and downward components for each layer, these results can be extended to the EM fields propagating in a layered earth if the spherical representation is enclosed by the layer under consideration.

This last point derives for the mathematical representation of the physics involved. The incident field is developed as a mathematical representation valid layer by layer. The spherical fields are developed from reflection coefficients assuming one uniform medium surrounding the sphere. Finally, the spherical to cylindrical field transformation is only valid in a uniform medium. If the sphere is in contact with a layer boundary, it is in contact with two mediums and the mathematical model is no longer valid. What should be the distance between the top or bottom of the sphere and the top or bottom of the layer has not been investigated, but it is assumed that as long as the tops or bottoms are numerically distinct, the model is valid.

6. Application of the reflection coefficients and layered earth reflection correction.
7. Representation of the spherical fields originating from the sphere as cylindrical expansions (Vallée and Moussaoui, 2023). One important aspect of the development of Gumerov and Duraiswami (2007) is that it allows the representation of the EM field Cartesian components as spherical expansions, whose coefficients are functions of the potential coefficients. The fields of particular interest are,

in the case of a layered earth, the vertical fields, described by equation (6) and (7). Furthermore, Vallée and Moussaoui (2023) demonstrate that these results can be extended to cylindrical functions and they even relate cylindrical and spherical functions. These algorithms have been applied in the following way in our program. First, from the sphere outward spherical potentials, we compute the spherical coefficients of the vertical EM fields at the equatorial plane of the sphere. Then, we develop a cylindrical expansion of these fields for a given wavenumber. Finally, in the wavenumber domain, we present these fields as vertical TM and TE potentials.

8. Propagation of the potential coefficients to the surface. The potentials are propagated to the surface. This includes the computation of the surface impedance and admittance (Ward and Hohmann, 1987), pp. 205-207 and the potential field coefficients. In Appendix E, the complete derivations of the cylindrical potential coefficients in a layered earth are presented. At the end of this computation, the field wave-number components are derived from the potentials by expressing derivatives as cylindrical expansions.
9. Transformation of the fields' cylindrical expansions at earth's surface using fast Hankel transform filters. The field components are transformed in the space domain using fast Hankel transform filters (Christensen, 1990; Raiche et al., 2007; Werthmüller, 2017).
10. Conversion of surface EM fields to MT measurements. The computation of magnetotelluric measurements is developed in the following way. The impedance is

computed from total magnetic and electric fields at surface. Apparent resistivity and phase are derived from the impedance.

Magnetotelluric modelling is simplified by the observation that plane wave fields can be represented by spherical unit of positive and negative order ($m = -1$ or $m = 1$). Furthermore, it can be demonstrated that propagation in the layered earth is independent of the order and that spherical potentials are related to vertical fields of the same order. In conclusion, the modelling can be done independently for each order, which simplifies the code and accelerates the solution.

The program is developed as a Fortran or Python program. The Fortran version has been compiled either with Intel or gfortran without any specific problem. Performance is good, for Intel or gfortran compilers, as a model with over 73 single frequency stations with 6 spherical terms can be run in less than a minute on a typical microcomputer. A description of the Fortran input is contained in a readme.txt file included with the distribution. The Python version accepts the same input file format or read the parameters for a menu. The sphere is located at the origin of the horizontal coordinates while MT measurements taken on the surface of the ground. The sphere must be located completely inside a layer and cannot touch the layer boundaries.

EXAMPLES

To illustrate the application of this program, we present a simple example of the magnetotelluric response of a conductive sphere of $1 \Omega \cdot \text{m}$ located at 100 m depth in a two-layered earth with a half-space resistivity of $1000 \Omega \cdot \text{m}$ below a layer of 30 m and $100 \Omega \cdot \text{m}$. Figure 2 presents a plan view and a section of the model with the geoelectrical parameters. A profile is conducted the X axis over the center of sphere while a profile along the Y axis is shifted on the side of the sphere. The soundings presented in Figure 3 are located at the intersection of the two profiles, between 1 mHz and 1 MHz, while the two profile models are simulated at 1 kHz. Figure 4 presents the apparent resistivity, phase and tipper for X profile and Figure 5 shows the same results for the Y profile.

COMPARISON WITH NUMERICAL RESULTS

To confirm the validity of this program, we compared its results with results of Finite Element (FE) and Finite Volume (FV) programs. Figures 6 and 7 show the FE designed model for a sphere with conductivity of $1.0 \Omega \cdot \text{m}$ and 50 m radius buried at 100 m in a homogeneous half-space with a conductivity value of $100 \Omega \cdot \text{m}$ and a frequency of 1 kHz. The number of cells in the model is 469,555. The apparent resistivity values of this model were computed using the edge-based FE approach (Jahandari and Farquharson, 2017; Kangazian and Farquharson, 2024) at a frequency of 1 kHz along the same X and Y profiles as in the previous section. The same profiles were simulated with a FV program developed following a demo (SimPEG

Developers (2026)) based on SimPEG framework (Heagy et al. (2017)). The mesh selected contains:

- $63 \times 63 \times 90 = 357210$ cells,
- $35 \times 35 \times 70$ core cells of $5 \times 5 \times 2.5$ m size,
- 14 padding cells in each direction with 1.4 exponential increment,
- the earth surface is set so that range between the minimum and maximum cell centers are -994.7 m and 829.7 m respectively.

This mesh is illustrated in Figures 8 and 9. Results from both programs are compared with the semi-analytic sphere response in Figure 10. FE and sphere responses are similar, but the FV response is either larger or smaller than the sphere on the slopes of the sphere anomaly.

Figures 11 and 12 show the FE designed model for a sphere with a resistivity of $1.0 \Omega \cdot \text{m}$ buried at 100 m from the surface in a half-space of $1000 \Omega \cdot \text{m}$ covered by a $100 \Omega \cdot \text{m}$ resistivity and 30 m thickness layer, as in the previous section. The number of cells in the FE model is 1,453,532. For the FV model, the discretization of the homogeneous earth model was also used for the layered earth model. The apparent resistivity and the tipper of this geology computed using FE, FV, and sphere programs are shown, respectively, in Figures 13 and 14. FE and sphere responses are very similar, but the FV solution shows a distinct response at large offsets. The three tipper responses are very similar.

The relative differences between FE, FV, and sphere responses are presented in Figures 15 and 16 as a percentage relative to the sphere response. The difference is larger for the half-space FV response which is either lower or higher than the sphere response on the slopes of the sphere anomaly, as observed before. For the FE response, the difference is below 2 %, particularly for the layered earth model. Note that the FV program was run on a 12th Gen Intel(R) Core(TM), 2.3 GHz with 16 Go RAM memory. It is expected that better results could be obtained with more computer resources.

CONCLUSION

We propose a new open source program, available in Fortran and Python, for modelling the magnetotelluric response of a sphere in a layered earth. This program was developed from a semi-analytic algorithm involving various mathematical tools assembled for this task. This program is of simple use and could be integrated in more complex programs, like modelling the response of two spheres. The primary application of this program might be for developers looking to validate numerical codes, as shown in the examples presented. Another application is the study of magnetotelluric fields around a discrete conductor in a layered environment.

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DATA AND MATERIAL AVAILABILITY

The Fortran and Python codes along with the examples presented in this paper, are open source and can be freely accessed as supplementary material through <https://github.com/marcavallee/mtsphere>. Comments or suggestions should be addressed to the corresponding author.

APPENDIX A

PLANE-WAVE FIELD PROPAGATION

The propagation of a plane wave field in a layered earth is described by Ward and Hohmann (1987), p. 194-202. The EM fields $E_x(z)$ and $H_y(z)$ propagating in the layer j with the electric field along the x axis can be decomposed in an upward and a downward component such as

$$E_x(z) = E_{xj}^+ e^{-ik_j(z-z_j)} + E_{xj}^- e^{ik_j(z-z_j)} \quad (\text{A-1})$$

and

$$\begin{aligned} H_y(z) &= \frac{k_j}{\omega\mu_0} (E_{xj}^+ e^{-ik_j(z-z_j)} - E_{xj}^- e^{ik_j(z-z_j)}) \\ &= \frac{1}{Z_j} (E_{xj}^+ e^{-ik_j(z-z_j)} - E_{xj}^- e^{ik_j(z-z_j)}) \end{aligned} \quad (\text{A-2})$$

where E_{xj}^+ and H_{yj}^+ are propagating downward, E_{xj}^- and H_{yj}^- are propagating upward, z is the elevation (positive downward), z_j is the elevation of the top of the layer j , and k_j and Z_j are the layer j propagation constant and impedance respectively

$$\begin{aligned} k_j^2 &= \mu_0\epsilon\omega^2 - i\mu_0\sigma_j\omega \\ &= -\hat{y}\hat{z} \end{aligned} \quad (\text{A-3})$$

$$\begin{aligned}
Z_j &= \frac{\omega\mu_0}{k_j} \\
&= \sqrt{\frac{\hat{z}}{\hat{y}}}
\end{aligned} \tag{A-4}$$

where μ is the magnetic permeability, ϵ is the electric permittivity, ω is the radial frequency, σ_n is the conductivity of the layer j , $i = \sqrt{-1}$, and $\hat{z} = i\mu\omega$ and $\hat{y} = \sigma + i\epsilon\omega$ are the impedivity and admittivity respectively, From these definitions, the fields at the top of the layer j are

$$\begin{aligned}
E_x(z_j) &= E_{xj}^+ + E_{xj}^- \\
H_y(z_j) &= \frac{1}{Z_j}(E_{xj}^+ - E_{xj}^-)
\end{aligned} \tag{A-5}$$

and we define the surface impedance as

$$\hat{Z}_j = \frac{E_x(z_j)}{H_y(z_j)} \tag{A-6}$$

Application of the boundary conditions (continuity of the tangential electric and magnetic fields) at the top of the layer $j + 1$ gives

$$\begin{aligned}
E_x(z_{j+1}) &= E_{xj}^+ e^{-ik_j h_j} + E_{xj}^- e^{ik_j h_j} \\
H_y(z_{j+1}) &= \frac{1}{Z_j}(E_{xj}^+ e^{-ik_j h_j} - E_{xj}^- e^{ik_j h_j}) \\
\hat{Z}_{j+1} &= \frac{E_x(z_{j+1})}{H_y(z_{j+1})} \\
&= Z_j \frac{E_{xj}^+ e^{-ik_j h_j} + E_{xj}^- e^{ik_j h_j}}{E_{xj}^+ e^{-ik_j h_j} - E_{xj}^- e^{ik_j h_j}}
\end{aligned} \tag{A-7}$$

from which

$$\begin{aligned}
E_{xj}^+ &= \frac{(\hat{Z}_{j+1} + Z_j)}{(\hat{Z}_{j+1} - Z_j)e^{-2ik_j h_j} + (\hat{Z}_{j+1} + Z_j)} E_x(z_j) \\
E_{xj}^- &= \frac{(\hat{Z}_{j+1} - Z_j)e^{-2ik_j h_j}}{(\hat{Z}_{j+1} - Z_j)e^{-2ik_j h_j} + (\hat{Z}_{j+1} + Z_j)} E_x(z_j)
\end{aligned} \tag{A-8}$$

and the surface impedance can be developed from a recurrence relation as

$$\hat{Z}_j = Z_j \frac{\hat{Z}_{j+1} + Z_j \tanh(ik_j h_j)}{Z_j + \hat{Z}_{j+1} \tanh(ik_j h_j)} \tag{A-9}$$

Using the relation

$$\hat{Z}_n = Z_n \quad (\text{A-10})$$

where n is the lower layer of the stratified earth, we use equation (A-9) to compute the surface impedance recursively from the bottom of the layered earth to the top. The downward and upward electric field coefficients are then computed starting with the following equation

$$E_{x0}^- = \frac{\hat{Z}_1 - Z_0}{\hat{Z}_1 + Z_0} E_{x0}^+ \quad (\text{A-11})$$

assuming as initial value $E_{x0}^+ = 1$. E_{xn}^+ and E_{xn}^- for $n \in [1, N]$ are computed using equations (A-7) and (A-8). The solution for the orthogonal $E_y i^+$ and $E_y i^-$ can be computed using the same relations.

APPENDIX B

SPHERICAL REPRESENTATION OF A PLANE WAVE FIELD

As shown by Ward and Hohmann (1987), pp. 292-296, a plane wave field can be represented as a sum of spherical functions. In spherical coordinates, electric and magnetic fields, defined at locations $\mathbf{r} = \mathbf{f}(r, \theta, \phi)$, can be derived from two scalar potentials: ψ_a , transverse magnetic (TM), and ψ_f , transverse electric (TE), defined by Ward and Hohmann (1987), p. 281 as

$$\mathbf{E} = -\nabla \times \mathbf{r} \psi_f + \frac{1}{\hat{y}} \nabla \times \nabla \times \mathbf{r} \psi_a \quad (\text{B-1})$$

$$\mathbf{H} = \nabla \times \mathbf{r}\psi_a + \frac{1}{\hat{z}} \nabla \times \nabla \times \mathbf{r}\psi_f \quad (\text{B-2})$$

where ψ_a and ψ_f are solutions of the Helmholtz equation. These solutions can be expressed in spherical coordinates as regular or inward $R_n^m(\mathbf{r})$ and singular or outward $S_n^m(\mathbf{r})$ functions. The regular functions are expressed as

$$R_n^m(\mathbf{r}) = j_n(kr)Y_n^m(\theta, \phi) \quad (\text{B-3})$$

and the singular functions are

$$S_n^m(\mathbf{r}) = h_n(kr)Y_n^m(\theta, \phi) \quad (\text{B-4})$$

where $j_n(kr)$ and $h_n(kr)$ are spherical Bessel and Hankel functions respectively where $Y_n^m(\theta, \phi)$ are the orthogonal spherical harmonics defined as

$$\begin{aligned} Y_n^m(\theta, \phi) &= (-1)^m \sqrt{\frac{2n+1}{4\pi} \frac{(n-|m|)!}{(n+|m|)!}} P_n^{|m|}(\cos \theta) e^{im\phi} \\ P_n^{|m|}(\mu) &= \frac{(-1)^{|m|}}{2^n n!} (1 - \mu^2)^{|m|/2} \frac{d^{|m|+n}}{d\mu^{|m|+n}} (\mu^2 - 1)^n \\ n &= 0, 1, 2, \dots; m = -n, \dots, n \end{aligned} \quad (\text{B-5})$$

Potentials can be represented as sums of spherical functions where ψ_i and ψ_o are the inward and outward TM or TE potentials

$$\begin{aligned} \psi_i &= \sum_{n=1}^{\infty} (\psi_i)_n^m R_n^m(\mathbf{r}) = \sum_{n=1}^{\infty} (\psi_i)_n^m j_n(kr) Y_n^m(\theta, \phi) \\ \psi_o &= \sum_{n=1}^{\infty} (\psi_o)_n^m S_n^m(\mathbf{r}) = \sum_{n=1}^{\infty} (\psi_o)_n^m h_n(kr) Y_n^m(\theta, \phi) \end{aligned} \quad (\text{B-6})$$

If we apply these representations to differential equations relating radial fields and potentials, such as

$$\begin{aligned} E_r &= \frac{1}{y} \left(\frac{\partial^2}{\partial r^2} + k^2 \right) r\psi_a \\ H_r &= \frac{1}{\hat{z}} \left(\frac{\partial^2}{\partial r^2} + k^2 \right) r\psi_f \end{aligned} \quad (\text{B-7})$$

we find

$$\begin{aligned}(\mathbf{r} \cdot \mathbf{E})_n^m &= \frac{n(n+1)}{\hat{y}} (\psi_a)_n^m \\(\mathbf{r} \cdot \mathbf{H})_n^m &= \frac{n(n+1)}{\hat{z}} (\psi_f)_n^m\end{aligned}\tag{B-8}$$

This implies that the potentials coefficients can be derived from the radial field coefficients. To achieve this objective, Ward and Hohmann (1987) proceed as follow. Starting with an incident plane wave field E_x^p propagating downward, developed as

$$E_x^p = E_{x0}^+ e^{-ikz} = E_{x0}^+ e^{-ik_{pr} \cos \theta} \tag{B-9}$$

where z is the elevation oriented downward, and r and θ are referenced to the origin. The radial component of the plane wave is

$$E_r^p = E_{x0}^+ \frac{\cos \phi}{ikr} \frac{\delta}{\delta \theta} (e^{-ikr \cos \theta}) \tag{B-10}$$

Using equation (10.1.47) from Abramowitz and Stegun (1965), which is

$$e^{ir \cos \theta} = \sum_{n=0}^{\infty} i^n (2n+1) j_n(r) P_n(\cos \theta) \tag{B-11}$$

we conclude that

$$E_r^p = \frac{E_{x0}^+ \cos \phi}{ik_p r} \sum_{n=0}^{\infty} (-i)^n (2n+1) j_n(k_p r) \frac{\delta}{\delta \theta} P_n(\cos \theta) \tag{B-12}$$

From definition (B-5), we can write that

$$P_n^1(\cos \theta) = \frac{dP_n(\cos \theta)}{d\theta} \tag{B-13}$$

and, since for $m > n$, $P_n^m = 0$,

$$E_r^p = \frac{-iE_{x0}^+ \cos \phi}{k_p r} \sum_{n=1}^{\infty} i^{-n} (2n+1) j_n(kr) P_n^1(\cos \theta) \tag{B-14}$$

Using the definitions (B-3) and (B-5) , we develop as

$$E_r^p = \frac{iE_{x0}^+}{k_p r} \sum_{n=1}^{\infty} i^{-n} \sqrt{\pi n(n+1)(2n+1)} [R_n^{-1} + R_n^1] \quad (\text{B-15})$$

Using relation (B-6), the transverse magnetic radial potential is obtained as

$$(\psi_a)_n^{m=-1,1} = \frac{i\dot{y}E_{x0}^+}{k_p} i^{-n} \sqrt{\frac{\pi(2n+1)}{n(n+1)}} \quad (\text{B-16})$$

With a similar development, we can derive the following relation for the magnetic radial field

$$H_r^p = \sin \phi \sin \theta \frac{E_{x0}^+}{Z_p} e^{-ikr \cos \theta} \quad (\text{B-17})$$

which leads to an equation, analogous to equation (B-14),

$$H_r^p = \frac{-iE_{x0}^+ \sin \phi}{k_p r} \sum_{n=1}^{\infty} i^{-n} (2n+1) j_n(kr) P_n^1(\cos \theta) \quad (\text{B-18})$$

or

$$H_r^p = \frac{E_{x0}^+ \sqrt{\pi}}{kr Z_p} \sum_{n=1}^{\infty} i^{-n} \sqrt{n(n+1)(2n+1)} [R_n^1 - R_n^{-1}] \quad (\text{B-19})$$

from which the transverse electric potential is

$$(\psi_f)_n^m = \frac{m\dot{z}E_{x0}^+}{kZ_p} i^{-n} \sqrt{\frac{\pi(2n+1)}{n(n+1)}} \quad (\text{B-20})$$

For the upward component, the development is similar except we do have to be careful about sign issues. Instead of equations (B-8) and (B-11), we write

$$E_r^p = -E_{x0}^- \frac{\cos \phi}{ik_p r} \frac{\delta}{\delta \theta} (e^{-ikr \cos \theta}) \quad (\text{B-21})$$

and consequently

$$E_r^p = -\frac{E_{x0}^- \cos \phi}{ik_p r} \sum_{n=0}^{\infty} i^n (2n+1) j_n(k_p r) \frac{\delta}{\delta \theta} P_n(\cos \theta) \quad (\text{B-22})$$

The TM potential becomes

$$(\psi_a)_n^{m=-1,1} = -\frac{i\hat{y}E_{x0}^+}{k_p} i^n \sqrt{\frac{\pi(2n+1)}{n(n+1)}} \quad (\text{B-23})$$

As the plane wave propagates upward, the magnetic field is

$$H_y^p = -\frac{E_x^p}{Z_p} e^{-ik_p r \cos \theta} \quad (\text{B-24})$$

and the TE potential is

$$(\psi_f)_n^m = \frac{m\hat{z}E_{x0}^+}{kZ_p} i^n \sqrt{\frac{\pi(2n+1)}{n(n+1)}} \quad (\text{B-25})$$

If $E_p = E_y$ and the plane wave is propagating downward, this can be described by

$$\begin{aligned} E_r^p &= \sin \phi \sin \theta E_{y0}^+ e^{-ik_p r \cos \theta} \\ H_r^p &= -\cos \phi \sin \theta \frac{E_{y0}^+}{Z_p} e^{-ik_p r \cos \theta} \end{aligned} \quad (\text{B-26})$$

The TM and TE potentials become respectively

$$\begin{aligned} (\psi_a)_n^m &= \frac{m\hat{y}_p E_{y0}^+}{k_p} i^{-n} \sqrt{\frac{\pi(2n+1)}{n(n+1)}} \\ (\psi_f)_n^{m=-1,1} &= -\frac{i\hat{z}_p E_{y0}^+}{k_p Z_p} i^{-n} \sqrt{\frac{\pi(2n+1)}{n(n+1)}} \end{aligned} \quad (\text{B-27})$$

Finally, if $E_p = E_y$ is propagating upward, the potentials become

$$\begin{aligned} (\psi_a)_n^m &= -\frac{m\hat{y}_p E_{y0}^+}{k_p} i^n \sqrt{\frac{\pi(2n+1)}{n(n+1)}} \\ (\psi_f)_n^{m=-1,1} &= -\frac{i\hat{z}_p E_{y0}^+}{k_p Z_p} i^n \sqrt{\frac{\pi(2n+1)}{n(n+1)}} \end{aligned} \quad (\text{B-28})$$

APPENDIX C

DEVELOPMENT OF SPHERICAL REFLECTION

COEFFICIENTS

Spherical reflection coefficients relate incident and reflected fields in the vicinity of a sphere. Developing equations (B-1) and (B-2), spherical EM fields are derived from

radial TM and TE potentials (Ward and Hohmann (1987), eq. A2.1.24). Satisfying the boundary conditions for the tangential TM electric and magnetic components gives, where the index l and s represent the layer and the sphere medium, respectively

$$\begin{aligned} E_{il\theta} + E_{ol\theta} &= E_{is\theta} \\ \frac{1}{\hat{y}_l r} \frac{\partial^2 r \psi_{ail}}{\partial r \partial \theta} + \frac{1}{\hat{y}_l r} \frac{\partial^2 r \psi_{aol}}{\partial r \partial \theta} &= \frac{1}{\hat{y}_s r} \frac{\partial^2 r \psi_{ais}}{\partial r \partial \theta} \end{aligned} \quad (\text{C-1})$$

The θ derivative is independent from the medium and can be removed. Furthermore, using equation (B-3), we can write

$$(\psi_{ail})_n^m \frac{1}{\hat{y}_l r} \frac{\partial r j_n(k_l r)}{\partial r} + (\psi_{aol})_n^m \frac{1}{\hat{y}_l r} \frac{\partial r h_n(k_l r)}{\partial r} = (\psi_{ais})_n^m \frac{1}{\hat{y}_s r} \frac{\partial r j_n(k_s r)}{\partial r} \quad (\text{C-2})$$

Developing the derivatives, with $r = r_s$ as the radius of the sphere,

$$\begin{aligned} (\psi_{ail})_n^m \hat{y}_s [j_n(k_l r_s) + r_s k_l j_n'(k_l r_s)] + (\psi_{aol})_n^m \hat{y}_s [h_n(k_l r_s) + k_l r_s h_n'(k_l r_s)] = \\ (\psi_{ais})_n^m [j_n(k_s r_s) + r_s k_s j_n'(k_s r_s)] \end{aligned} \quad (\text{C-3})$$

Similarly, if we apply the boundary condition on H_ϕ ,

$$\begin{aligned} H_{il\phi} + H_{ol\phi} &= H_{is\phi} \\ \frac{1}{r} \frac{\partial r \psi_{ail}}{\partial \theta} + \frac{1}{r} \frac{\partial r \psi_{aol}}{\partial \theta} &= \frac{1}{r} \frac{\partial r \psi_{ais}}{\partial \theta} \\ (\psi_{ail})_n^m j_n(k_l r_s) + (\psi_{aol})_n^m h_n(k_l r_s) &= (\psi_{ais})_n^m j_n(k_s r_s) \end{aligned} \quad (\text{C-4})$$

If we eliminate $\psi_{ais})_n^m$ from equations (C-5) and (C-6) and reorganize, we find that

$$\begin{aligned} (R_{TM})_n^m &= \frac{\psi_{aol}}{\psi_{ail}} \\ &= -\frac{\hat{y}_s j_n(k_s r_s) [h_n(k_l r_s) + k_l r_s h_n'(k_l r_s)] - \hat{y}_l h_n(k_l r_s) [j_n(k_s r_s) + k_s r_s j_n'(k_s r_s)]}{\hat{y}_s j_n(k_s r_s) [j_n(k_l r_s) + k_l r_s j_n'(k_l r_s)] - \hat{y}_l j_n(k_l r_s) [j_n(k_s r_s) + k_s r_s j_n'(k_s r_s)]} \end{aligned} \quad (\text{C-5})$$

Similarly, the RE reflection coefficient is

$$\begin{aligned} (R_{TM})_n^m &= \frac{\psi_{fol}}{\psi_{fil}} \\ &= -\frac{\hat{z}_s j_n(k_s r_s) [h_n(k_l r_s) + k_l r_s h_n'(k_l r_s)] - \hat{z}_l h_n(k_l r_s) [j_n(k_s r_s) + k_s r_s j_n'(k_s r_s)]}{\hat{z}_s j_n(k_s r_s) [j_n(k_l r_s) + k_l r_s j_n'(k_l r_s)] - \hat{z}_l j_n(k_l r_s) [j_n(k_s r_s) + k_s r_s j_n'(k_s r_s)]} \end{aligned} \quad (\text{C-6})$$

APPENDIX D

CYLINDRICAL EXPANSIONS

Relations similar to the developments described by equations (1) to (3) can be proposed for cylindrical waves and lead to relations with spherical functions. EM cylindrical waves can be represented using the following generalized Hankel transforms

$$\sum_{m=-\infty}^{\infty} I_m(\rho, \phi, z) = \sum_{m=-\infty}^{\infty} \int_0^{\infty} \lambda J_{|m|}(\lambda \rho) e^{-u|z|} e^{im\phi} d\lambda \quad (\text{D-1})$$

or

$$I_m(\rho, \phi, z) = \int_0^{\infty} \lambda G_m(\rho, \lambda, \phi, z) d\lambda \quad (\text{D-2})$$

where λ is the Hankel transform wavenumber, $u = \sqrt{\lambda^2 - k^2}$, and m is the order.

As with spherical functions, we can develop derivative expansions of these functions

which are

$$\begin{aligned} \left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)G_m(\rho, \lambda, \phi, z) &= \begin{cases} -\text{sgn}(m)\lambda G_{m+1}(\rho, \lambda, \phi, z), & m \neq 0 \\ -\lambda G_1(\rho, \lambda, \phi, z), & m = 0 \end{cases} \\ \left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)G_m(\rho, \lambda, \phi, z) &= \begin{cases} \text{sgn}(m)\lambda G_{m-1}(\rho, \lambda, \phi, z), & m \neq 0 \\ -\lambda G_{-1}(\rho, \lambda, \phi, z), & m = 0 \end{cases} \\ \frac{\partial}{\partial z}G_m(\rho, \lambda, \phi, z) &= -\text{sgn}(z)uG_m(\rho, \lambda, \phi, z) \end{aligned} \quad (\text{D-3})$$

Cylindrical coefficient expansions can be developed from these relations. If we assume that

$$\sum_{m=-\infty}^{\infty} \hat{\chi}_m G_m(\rho, \lambda, \phi, z) = \frac{\partial}{\partial w} \sum_{m'=-\infty}^{\infty} \chi_{m'} G_{m'}(\rho, \lambda, \phi, z) \quad (\text{D-4})$$

from equation (D-13) we derive the following solution

$$\hat{\chi}_m = \sum_{m'=-\infty}^{\infty} (D_w)_{mm'} \chi_{m'} \quad (\text{D-5})$$

where

$$\begin{aligned} (D_{x+iy})_{mm'} &= \begin{cases} -\delta_{m,m'+1} \text{sgn}(m') \lambda \chi_{m'} & m' \neq 0 \\ -\delta_{m,m'+1} \lambda \chi_{m'}, & m' = 0 \end{cases} \\ (D_{x-iy})_{mm'} &= \begin{cases} \delta_{m,m'-1} \text{sgn}(m') \lambda \chi_{m'}, & m' \neq 0 \\ -\delta_{m,m'-1} \lambda \chi_{m'}, & m' = 0 \end{cases} \\ (D_z)_{mm'} &= -\delta_{m,m'} \text{sgn}(z) u \chi_{m'} \end{aligned} \quad (\text{D-6})$$

Using these relations, EM fields can be represented as sums of cylindrical functions derived from potential coefficients associated with the different sources. Equations (D-6) are applied successfully in relations (A1.1.22) of Ward and Hohmann (1987) to compute the field components in the wavenumber domain from the vertical potentials. Furthermore, using derivative relations, we can express singular spherical functions as cylindrical expansions. Spherical and cylindrical functions are related through the Sommerfeld (1909) integral which describes the field originating from a point source as a Hankel transform

$$\frac{e^{-ik}}{r} = \int_0^{\infty} \frac{\lambda}{u} e^{-u|z|} J_0(\lambda \rho) d\lambda \quad (\text{D-7})$$

where the left side represents a spherical function and the right side represents an Hankel transform. This Hankel transform can be generalized as a cylindrical function by seeking solutions which are a summation of cylindrical functions of order m such as

$$S_n^m(\mathbf{r}) = \sum_{m'=-\infty}^{\infty} \int_0^{\infty} \zeta_{nm'}^m G_{m'}(\rho, \lambda, \phi, z) \lambda d\lambda \quad (\text{D-8})$$

ζ_{nm}^m is evaluated in the following way. First, the following recurrent relations can be derived for equations (1) to (3).

$$S_n^m(\mathbf{r}) = \begin{cases} \frac{1}{kb_n^m} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) S_{n-1}^{m-1}(\mathbf{r}), & m = n \\ \frac{1}{kb_n^m} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) S_{n-1}^{m+1}(\mathbf{r}), & m = -n \\ \frac{a_{n-2}^m}{a_{n-1}^m} S_{n-2}^m(\mathbf{r}) - \frac{1}{ka_{n-1}^m} \frac{\partial}{\partial z} S_{n-1}^m(\mathbf{r}), & -n < m < n \end{cases} \quad (\text{D-9})$$

The recurrence start is derived from the Sommerfeld integral (D-7) using the definitions of the spherical singular functions (B-4) and (B-6)

$$S_0^0(\mathbf{r}) = \int_0^\infty \frac{i}{ku\sqrt{4\pi}} \lambda G_0(\rho, \lambda, z) d\lambda \quad (\text{D-10})$$

Then, the sparse matrix ζ_{nm}^m can be developed from relations (D-6) and (D-9).

APPENDIX E

LAYERED EARTH PROPAGATION

We develop propagation parameters of TM and TE fields following Ward and Hohmann (1987), which provide the formula for computing the surface impedance and admittance, while we add new developments for computing the propagation coefficients at each interface of the layered earth. The boundaries conditions are continuity of the electric and magnetic field. As presented by Ward and Hohmann (1987), we use vertical potentials to describe propagation in the layered earth because derivatives can be evaluated using cylindrical expansions using the algorithm described in Appendix D. Propagation from the sphere to any location in the layered earth can be represented by the following equations in the wavenumber domain evaluating the potentials A_z

and F_z

$$\begin{aligned} A_z &= A_s^m A_n \\ F_z &= F_s^m F_n \end{aligned} \tag{E-1}$$

where A_s^m and F_s^m are the sphere TM and TE potentials, computed with the algorithm presented in Appendix D, and A_n and F_n are the TM and TE propagation factors. Note that, because of the continuity of the electric and magnetic fields at the horizontal layer boundaries, the propagation factors are independent of the cylindrical order. Then it is only necessary, in the wavenumber domain, to develop a propagation algorithm that is valid for all orders.

In source free layers, the potential of upward and downward propagating waves can be represented by

$$A_n = A_n^+ e^{-u_n(z-z_n)} + A_n^- e^{u_n(z-z_n)} \tag{E-2}$$

At the interface $z = z_n$, the surface impedance is, applying $\mathcal{Z}_n = u_n/\hat{y}_n$,

$$\hat{\mathcal{Z}}_n = \frac{\mathcal{Z}_n A_n^+ - \mathcal{Z}_n A_n^-}{A_n^+ + A_n^-} \tag{E-3}$$

Similarly at the layer $z = z_{n+1}$, where $h_n = z_{n+1} - z_n$,

$$\hat{\mathcal{Z}}_{n+1} = \frac{\mathcal{Z}_n A_n^+ e^{-u_n h_n} - \mathcal{Z}_n A_n^- e^{u_n h_n}}{A_n^+ e^{-u_n h_n} + A_n^- e^{u_n h_n}} \tag{E-4}$$

Developing and simplifying, where , we find that

$$\hat{\mathcal{Z}}_n = \mathcal{Z}_n \frac{\hat{\mathcal{Z}}_{n+1} + \mathcal{Z}_n \tanh(u_n h_n)}{\mathcal{Z}_n + \hat{\mathcal{Z}}_{n+1} \tanh(u_n h_n)} \tag{E-5}$$

and

$$\hat{\mathcal{Z}}_{n+1} = \mathcal{Z}_n \frac{\hat{\mathcal{Z}}_n - \mathcal{Z}_n \tanh(u_n h_n)}{\mathcal{Z}_n - \hat{\mathcal{Z}}_n \tanh(u_n h_n)} \tag{E-6}$$

For $\hat{\mathcal{Z}}_1$ and $\hat{\mathcal{Z}}_N$, the bottom and the top of the layered earth, if these half-spaces are source free,

$$\begin{aligned}\hat{\mathcal{Z}}_1 &= -\mathcal{Z}_0 \\ \hat{\mathcal{Z}}_N &= \mathcal{Z}_N\end{aligned}\tag{E-7}$$

Using equations (E-7) to start the recursion, and equations (E-5) recursively or (E-6) forward, we can compute the surface impedances for all the layers, if there is only one source.

These results are a first step in estimating the propagation coefficients. The first step is to consider the sphere layer $z = z_p$, which contains the sphere centre located at $z = z_s$, for which the propagation coefficients are

$$A_p = A_p^+ e^{-u_p(z-z_p)} + A_p^- e^{u_p(z-z_p)} + A_s e^{-|u_p(z-z_s)|}\tag{E-8}$$

where $A_s = A_s^-$ for $z < z_s$ and $A_s = A_s^+$ for $z > z_s$. At $z = z_p$, the surface impedance is

$$\hat{\mathcal{Z}}_p = -\frac{-\mathcal{Z}_p A_p^+ + \mathcal{Z}_p A_p^- + \mathcal{Z}_p A_s^- e^{-u_p(z_s-z_p)}}{A_p^+ + A_p^- + A_s^- e^{-u_p(z_s-z_p)}}\tag{E-9}$$

Rearranging

$$(\hat{\mathcal{Z}}_p - \mathcal{Z}_p)A_p^+ + (\hat{\mathcal{Z}}_p + \mathcal{Z}_p)A_p^- = -(\hat{\mathcal{Z}}_p + \mathcal{Z}_p)e^{-u_p(z_s-z_p)}A_s^-\tag{E-10}$$

Similarly, for $z = z_{p+1}$,

$$(\hat{\mathcal{Z}}_{p+1} - \mathcal{Z}_p)e^{-2u_p h_p} A_p^+ + (\hat{\mathcal{Z}}_{p+1} + \mathcal{Z}_p)A_p^- = -(\hat{\mathcal{Z}}_{p+1} - \mathcal{Z}_p)e^{-u_p[(z_{p+1}-z_s)+h_p]}A_s^+\tag{E-11}$$

Solving (E-10) and (E-11), we find that the propagation coefficients are

$$\begin{aligned} A_p^+ &= -(\hat{\mathcal{Z}}_p + \mathcal{Z}_p) \frac{(\hat{\mathcal{Z}}_{p+1} + \mathcal{Z}_p) e^{-u_p(z_s - z_p)} A_s^- - (\hat{\mathcal{Z}}_{p+1} - \mathcal{Z}_p) e^{-u_p h_p} e^{-u_p(z_{p+1} - z_s)} A_s^+}{(\hat{\mathcal{Z}}_p - \mathcal{Z}_p)(\hat{\mathcal{Z}}_{p+1} + \mathcal{Z}_p) - (\hat{\mathcal{Z}}_p + \mathcal{Z}_p)(\hat{\mathcal{Z}}_{p+1} - \mathcal{Z}_p) e^{-2u_p h_p}} \\ A_p^- &= (\hat{\mathcal{Z}}_{p+1} - \mathcal{Z}_p) \frac{(\hat{\mathcal{Z}}_p + \mathcal{Z}_p) e^{-2u_p h_p} e^{-u_p(z_s - z_p)} A_s^- - (\hat{\mathcal{Z}}_p - \mathcal{Z}_p) e^{-u_p h_p} e^{-u_p(z_{p+1} - z_s)} A_s^+}{(\hat{\mathcal{Z}}_p - \mathcal{Z}_p)(\hat{\mathcal{Z}}_{p+1} + \mathcal{Z}_p) - (\hat{\mathcal{Z}}_p + \mathcal{Z}_p)(\hat{\mathcal{Z}}_{p+1} - \mathcal{Z}_p) e^{-2u_p h_p}} \end{aligned} \quad (\text{E-12})$$

If the source is located in the bottom layer, where $A_z = A_N$, we find

$$A_N = A_N^+ e^{-u_N(z - z_N)} + A_s^- e^{-|u_N(z - z_s)|} \quad (\text{E-13})$$

the impedance $\hat{\mathcal{Z}}_N$ is

$$\hat{\mathcal{Z}}_N = \frac{\mathcal{Z}_N A_N^+ - \mathcal{Z}_N A_s^- e^{u_N(z_N - z_s)}}{A_N^+ + A_s^- e^{u_N(z_N - z_s)}} \quad (\text{E-14})$$

Rearranging, we find

$$A_N^+ = \frac{\mathcal{Z}_N + \hat{\mathcal{Z}}_N}{\mathcal{Z}_N - \hat{\mathcal{Z}}_N} e^{-u_N(z_s - z_N)} A_s^- \quad (\text{E-15})$$

For the field in source free layers, we compute the potentials in a layer from the potential of the layer above or below. First, if we know the potential at the surface n , we derive the upward and downward components in the following way. From equation (E-2) at the interface $z = z_n$ and equation (E-10) for a source free layer, we develop

$$\begin{aligned} A_n^+ &= \frac{\mathcal{Z}_n + \hat{\mathcal{Z}}_n}{2\hat{\mathcal{Z}}_n} A_n \\ A_n^- &= \frac{\mathcal{Z}_n - \hat{\mathcal{Z}}_n}{2\hat{\mathcal{Z}}_n} A_n \end{aligned} \quad (\text{E-16})$$

Then, these expressions are used in equation (E-2) for $z = z_{n+1}$ to develop the following relation

$$A_{n+1} = \frac{\mathcal{Z}_n(e^{-u_n h_n} + e^{u_n h_n}) + \hat{\mathcal{Z}}_n(e^{-u_n h_n} - e^{u_n h_n})}{2\hat{\mathcal{Z}}_n} A_n \quad (\text{E-17})$$

which allow the computation of the potential at surfaces above the sphere or using the inverse of equation (E-17) for surfaces below the sphere such as:

$$A_n = \frac{2\hat{\mathcal{Z}}_n e^{-u_n h_n}}{\mathcal{Z}_n(1 + e^{-2u_n h_n}) - \hat{\mathcal{Z}}_n(1 - e^{-2u_n h_n} - 1)} A_{n+1} \quad (\text{E-18})$$

At the top of the layers, there is only an upward component and

$$A_1 = A_0^- \quad (\text{E-19})$$

Similarly, at the bottom layer, if the bottom layer is source free, we can write

$$A_N = A_N^+ \quad (\text{E-20})$$

The TE propagation can be developed by replacing the TM potential A_z by the TE potential F_z and the impedance $\mathcal{Z}$ by the admittance $\mathcal{Y} = u/\hat{z}$.

APPENDIX F

SPHERE SELF-INTERACTION

The field originating from the sphere is reflected by the layer boundaries. This reflection is a function of the spherical harmonics and can be estimated if we represent spherical potentials by vectors as such. Since propagation of a different order is independent, we can represent the potentials of a specific order $m = -1$ or $m = 1$ as a vector:

$$v = \begin{bmatrix} (\psi_a)_1^m \\ (\psi_a)_2^m \\ \dots \\ (\psi_f)_1^m \\ \dots \end{bmatrix} \quad (\text{F-1})$$

where $(\psi_a)_n^m$ and $(\psi_f)_n^m$ are the coefficients of the radial TM and TE potentials. As the incident field on the sphere is a sum on the field from the EM source and the field reflected by the layered earth, and as the relation between the potentials is linear, we can write

$$v_i^m = v_s^m + v_r^m \quad (\text{F-2})$$

where:

- v_i^m : potential vector incident on the sphere for order m ,
- v_s^m : potential vector from the EM source for order m ,
- v_r^m : reflected potential vector for order m .

where the reflected potential vector v_r^m is reflected by the layered earth on the sphere for order m . Introducing the layered earth reflection as a matrix operation,

$$v_r^m = C^m v_i^m = v_i^m - v_s^m \quad (\text{F-3})$$

where C^m is a square matrix where each column is the response of one line on the incident potential vector. This matrix is only a function of the electrical parameters of the sphere and the layered earth and can be evaluated independently of the primary field and the receiver locations by computing separately the reflected potential to each incident potential. Consequently, we can write

$$v_i^m - C^m v_i^m = v_s^m \quad (\text{F-4})$$

$$v_i^m = (I - C^m)^{-1} v_s^m \quad (\text{F-5})$$

This last equation relates sphere incident fields to source fields and allows the correction of the source primary field for the layered earth reflection. In practice, this requires computing the reflected field on the sphere and estimating the reflected potential. This is done by estimating the radial EM fields reflected on the sphere, and computing spherical harmonics of the fields using spherical harmonic analysis.

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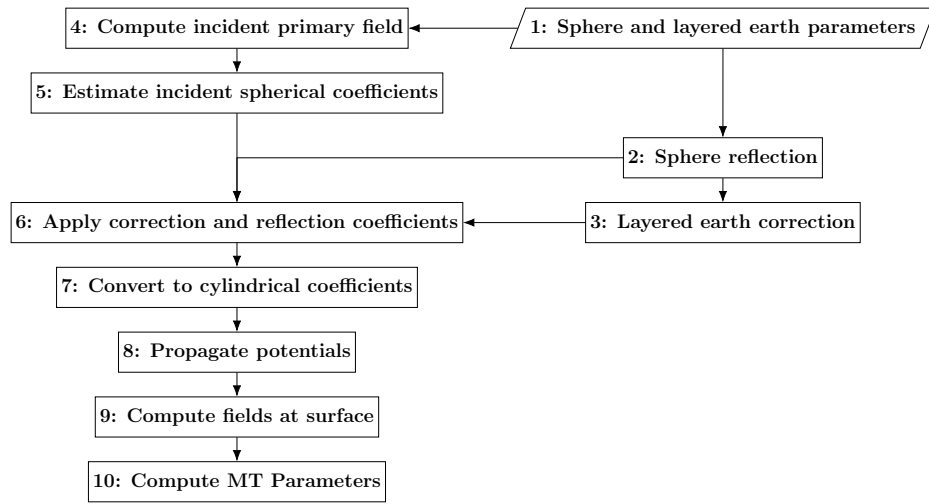
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Figure 1: Flow chart depicting the modelling algorithm. Each number refers to a step in the process which is then explained further in the text.

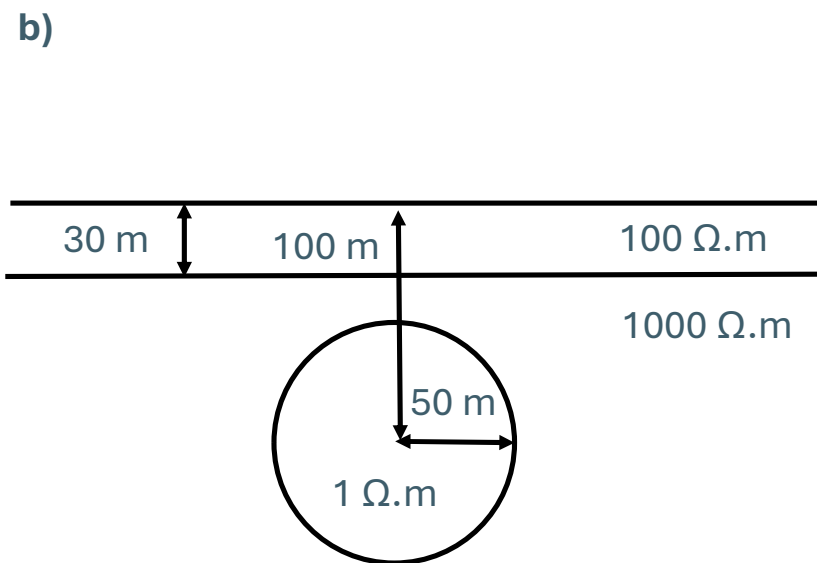
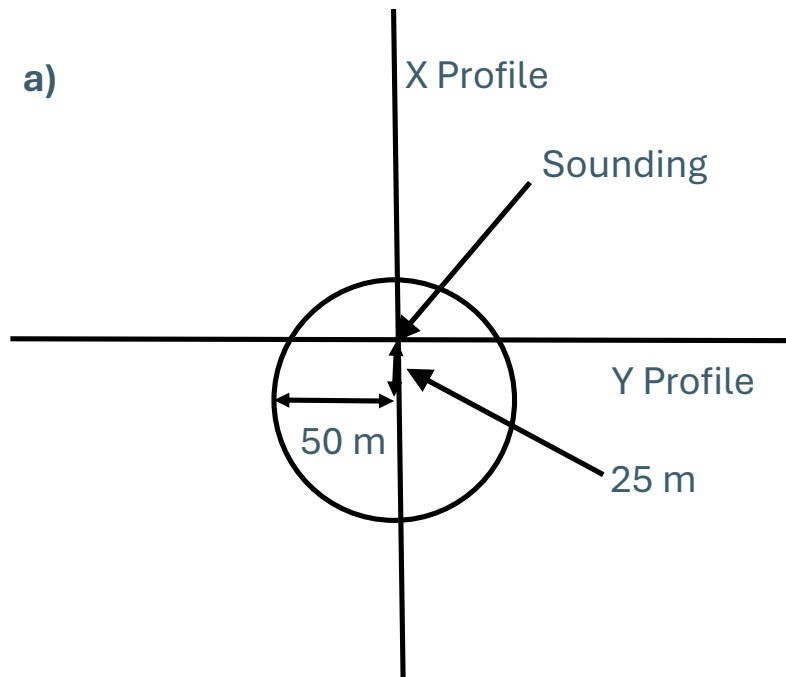


Figure 2: (a) Plan view and (b) section of the model whose results are presented.

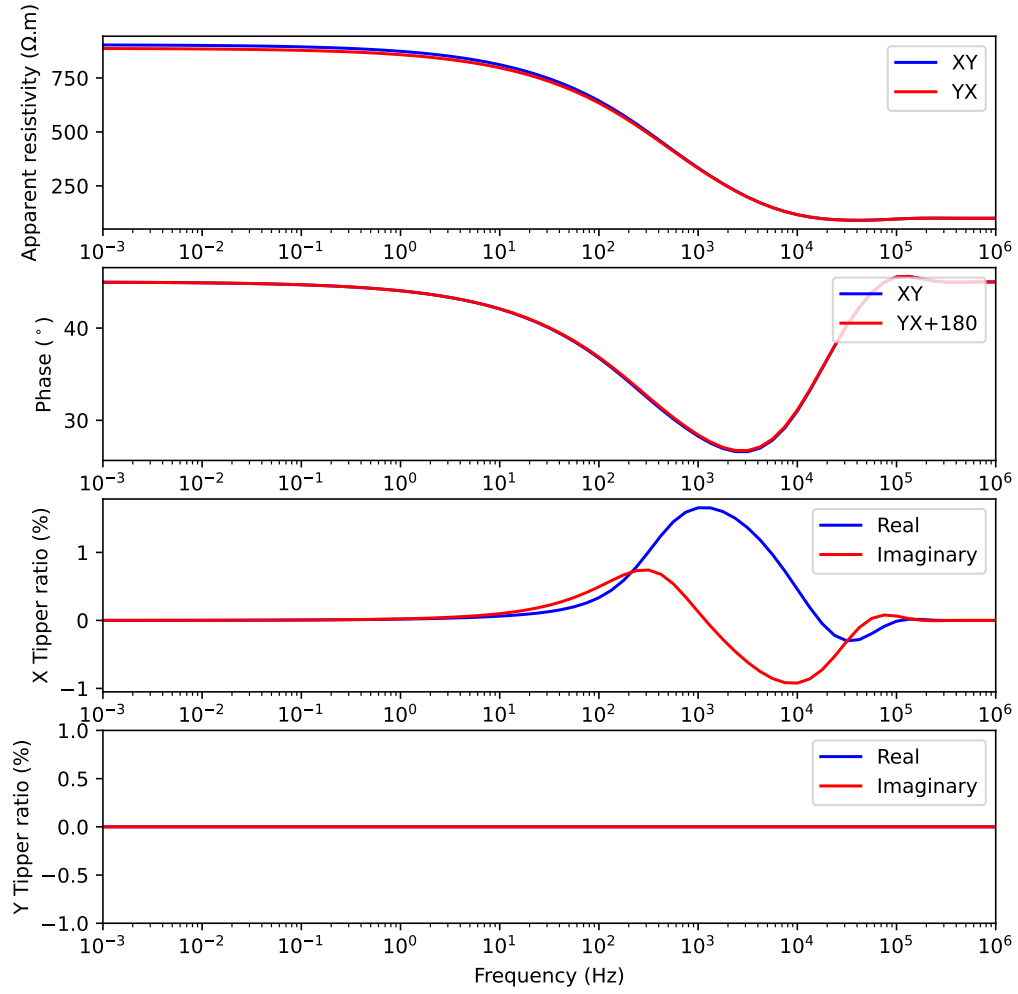


Figure 3: Sounding apparent resistivity, phase, and X and Y oriented Tippers for the location shown in Figure 2.

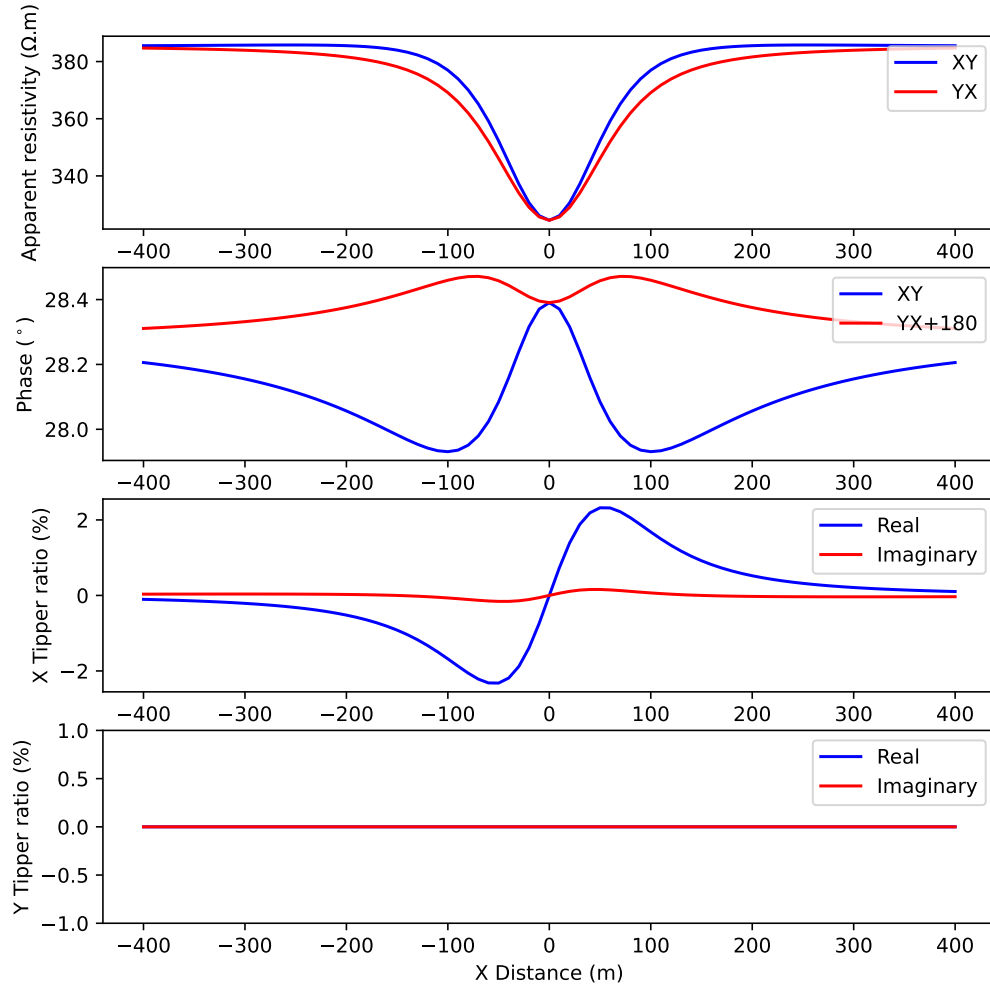


Figure 4: Apparent resistivity, phase, and X and Y oriented Tippers along the X profile.

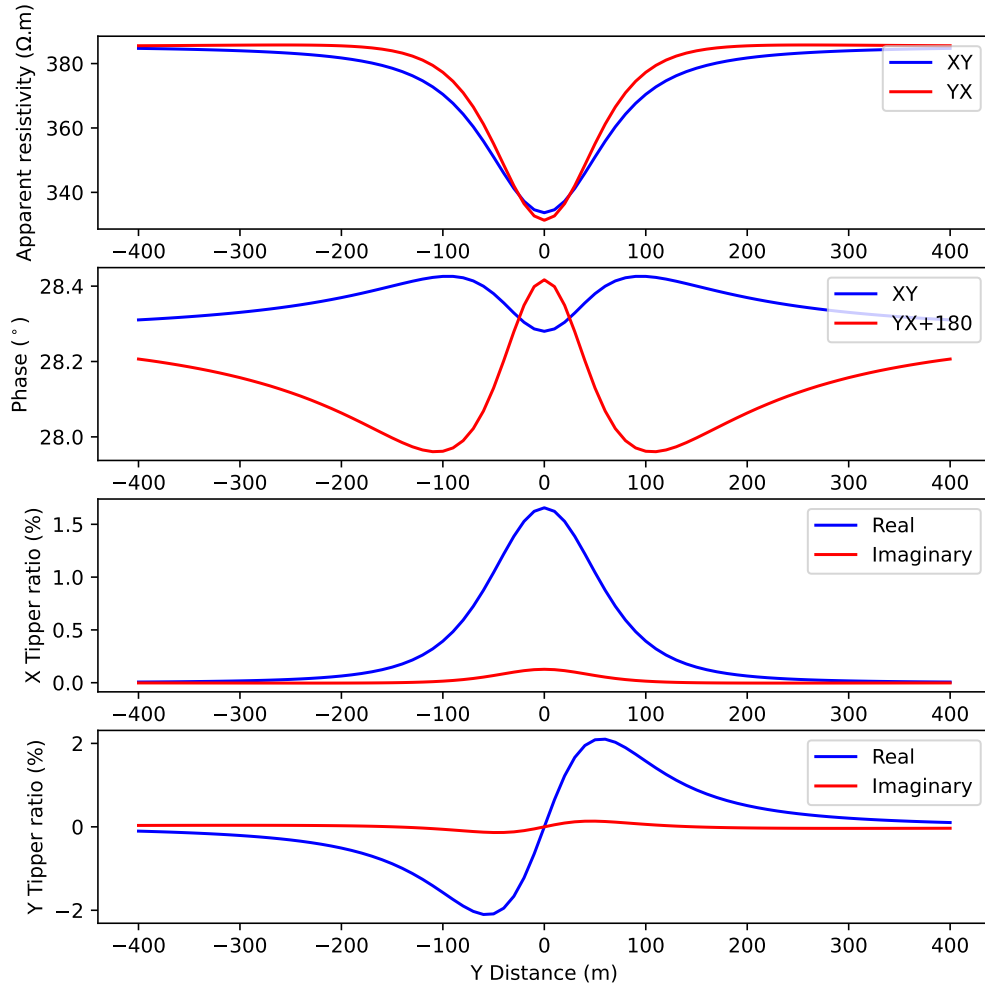


Figure 5: Apparent resistivity, phase, and X and Y oriented Tippers along the Y profile.

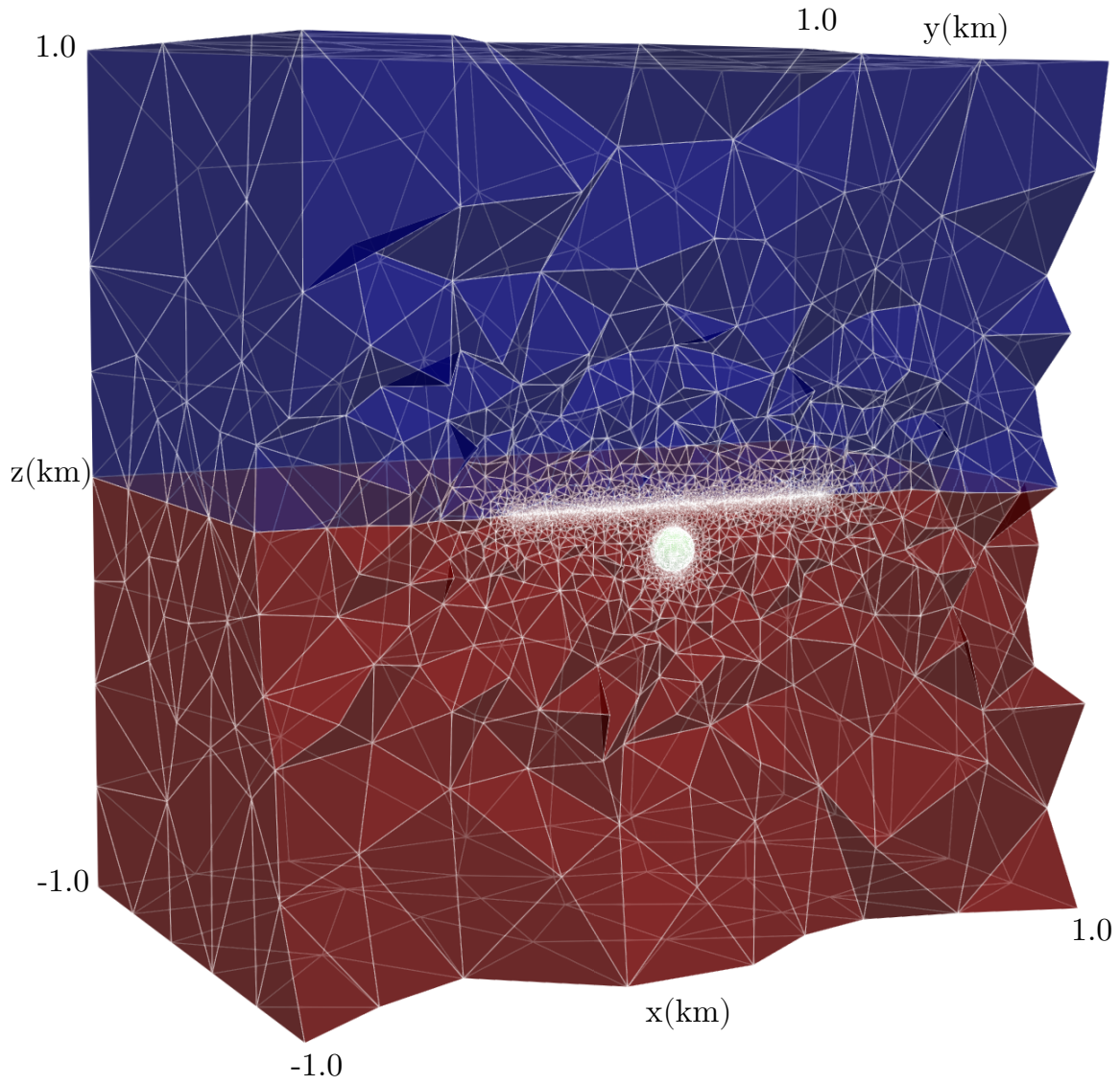


Figure 6: Forward modelling mesh used by the FE approach to calculate the electric and magnetic fields, impedance tensor data, apparent resistivity, and phase values of a sphere buried in a homogeneous half-space.

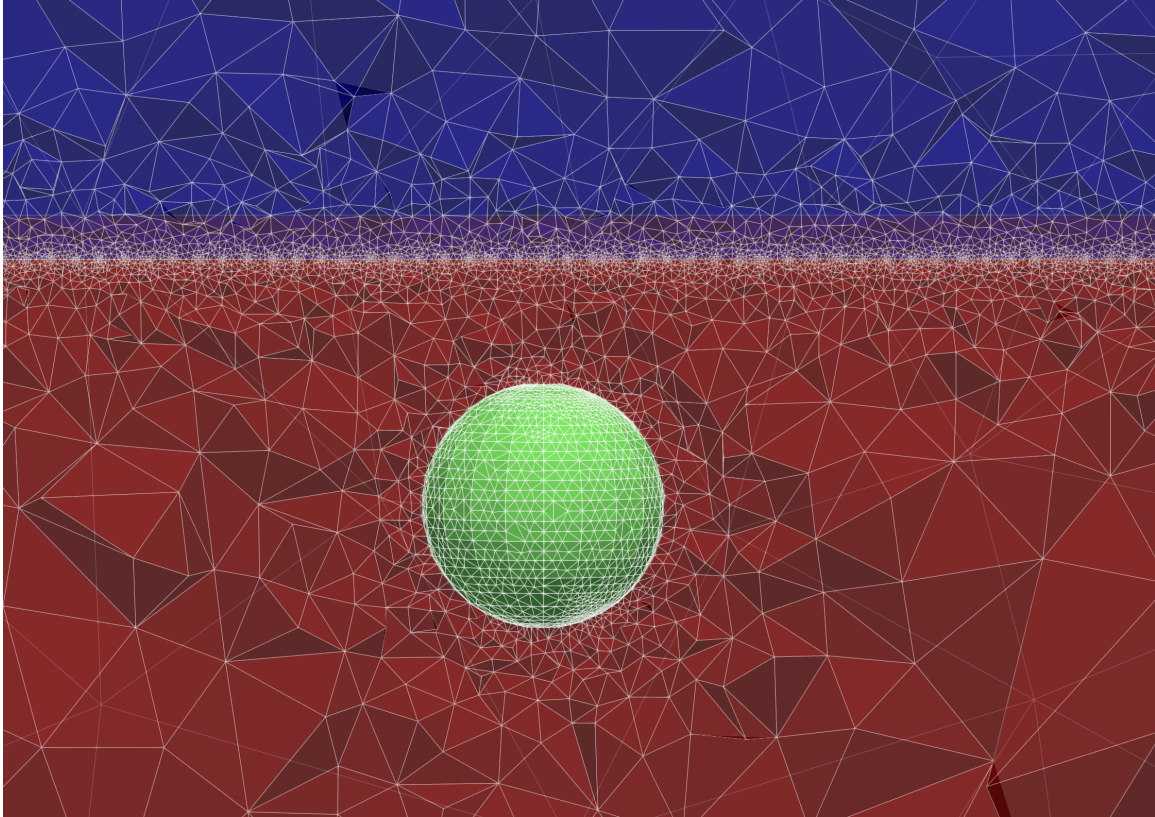


Figure 7: Closeup of the forward modelling mesh used by the FE approach to calculate the electric and magnetic fields, impedance tensor data, apparent resistivity, and phase values of a sphere buried in a homogeneous half-space.

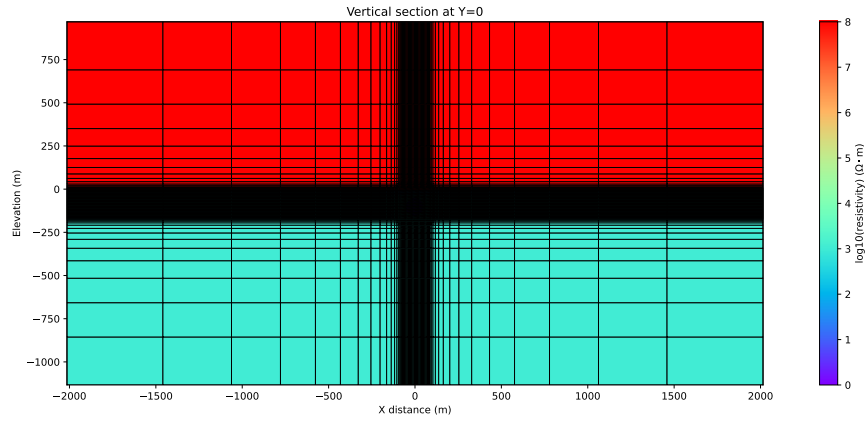


Figure 8: Forward modelling mesh used by the FV approach to calculate the electric and magnetic fields, impedance tensor data, apparent resistivity, and phase values of a sphere buried in a homogeneous half-space. The same discretization was used for the homogeneous and layered earth models.

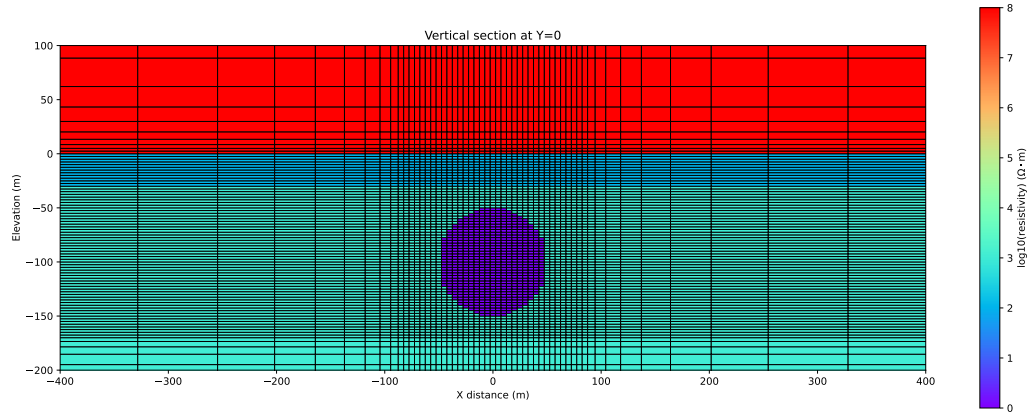


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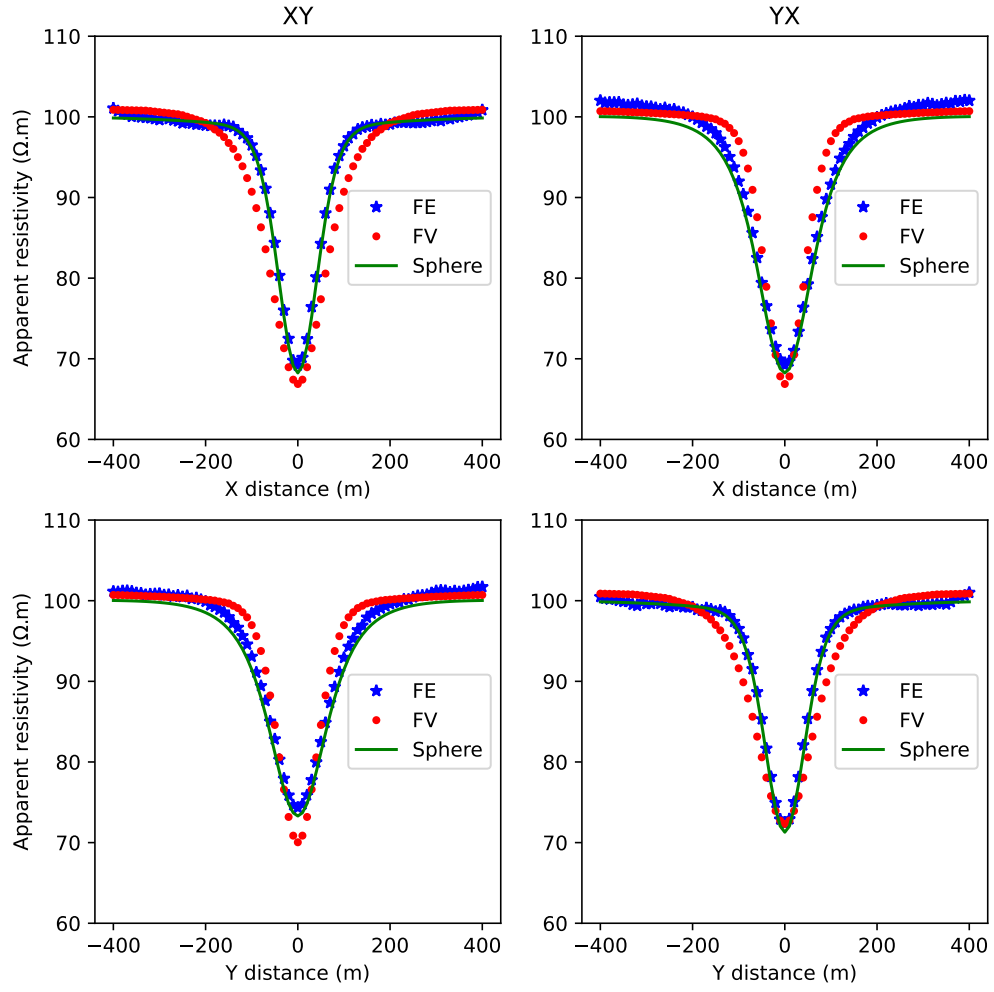


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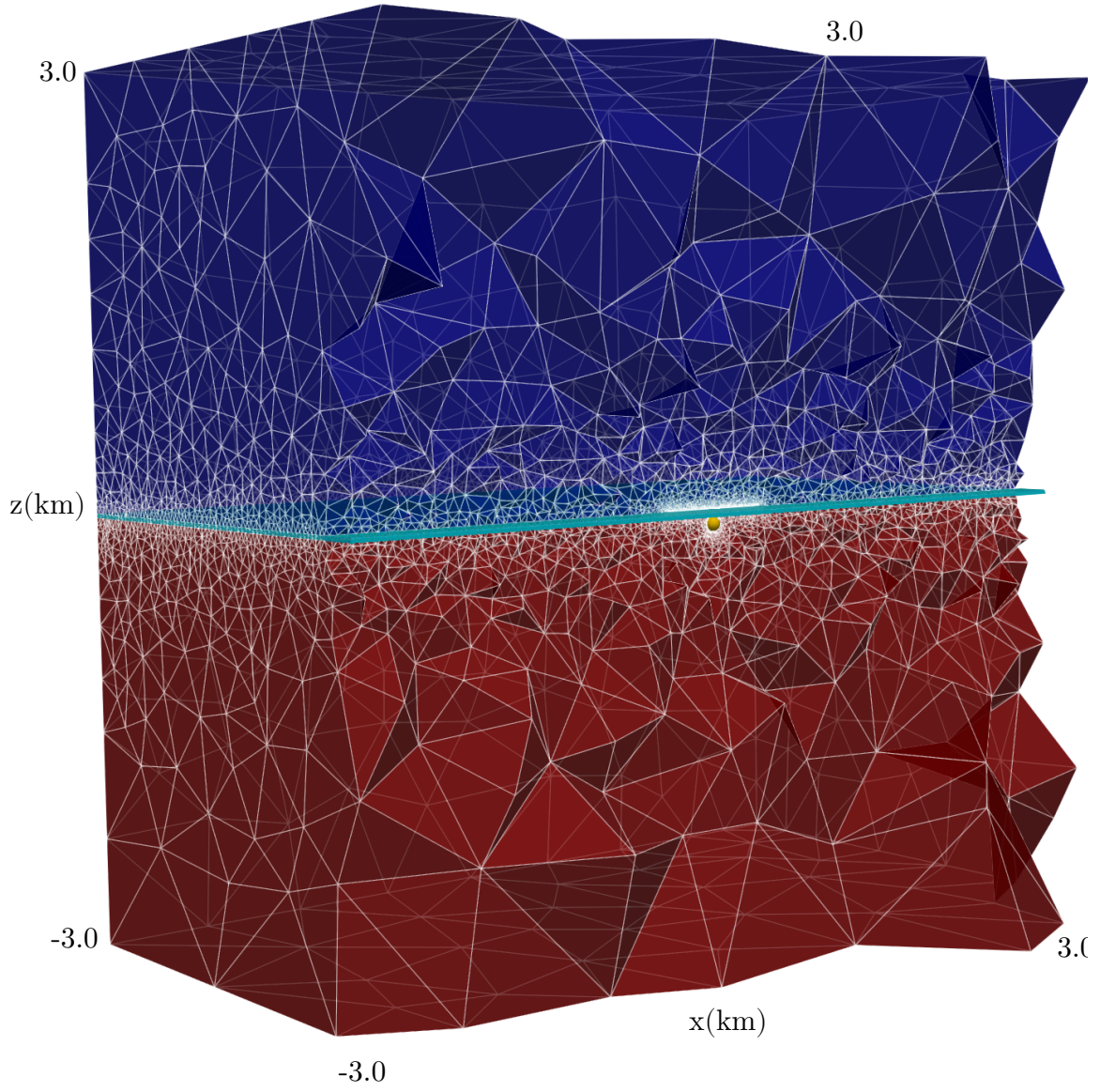


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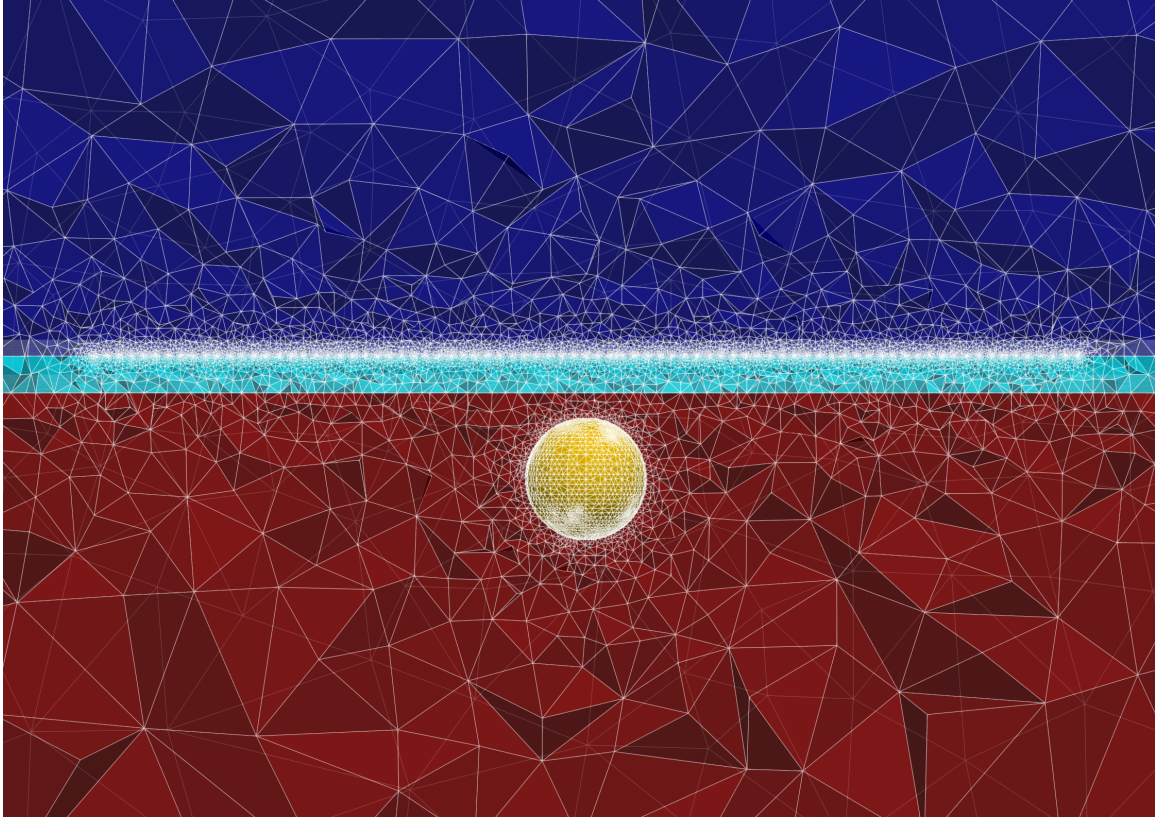


Figure 12: Closeup of the forward modelling mesh used by the FE approach to calculate the electric and magnetic fields, impedance tensor data, apparent resistivity, and phase values of a sphere buried in a layered-earth model.

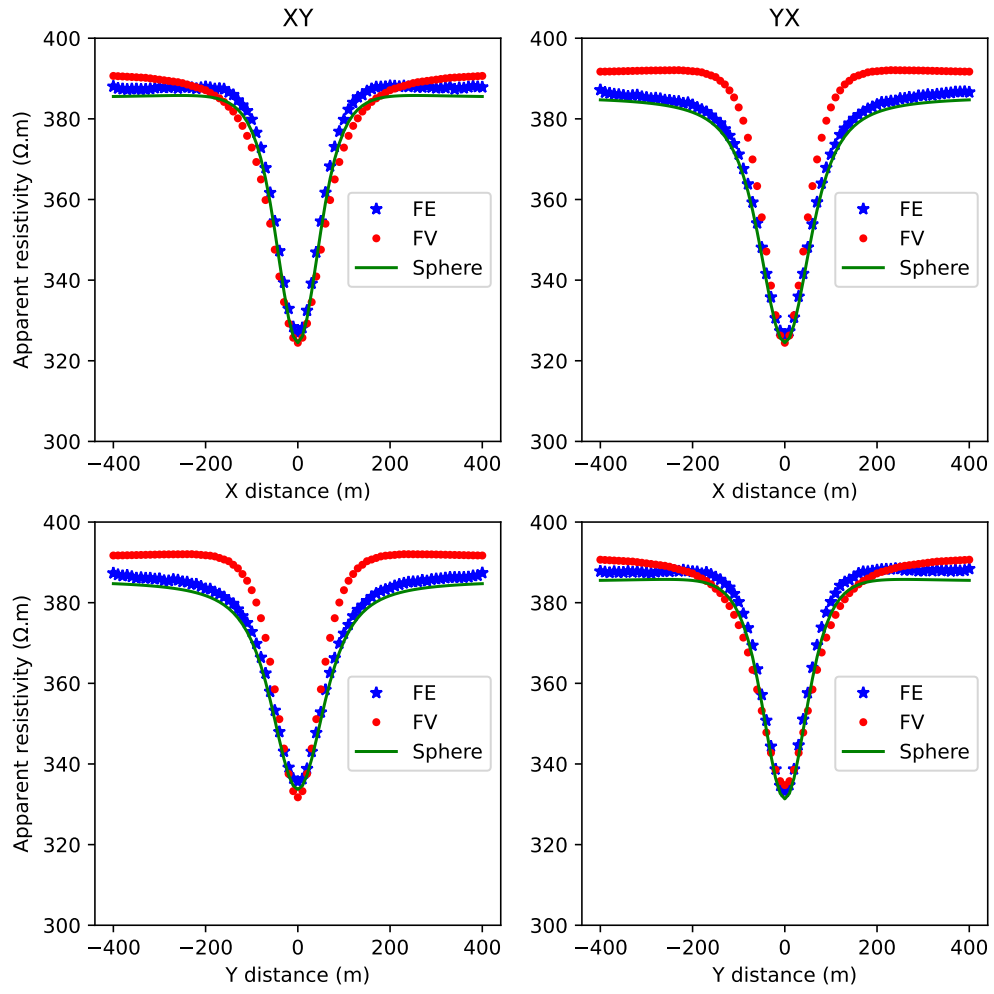


Figure 13: Apparent resistivity response of a sphere buried in a layered-earth model for sphere, FE, and FV models.

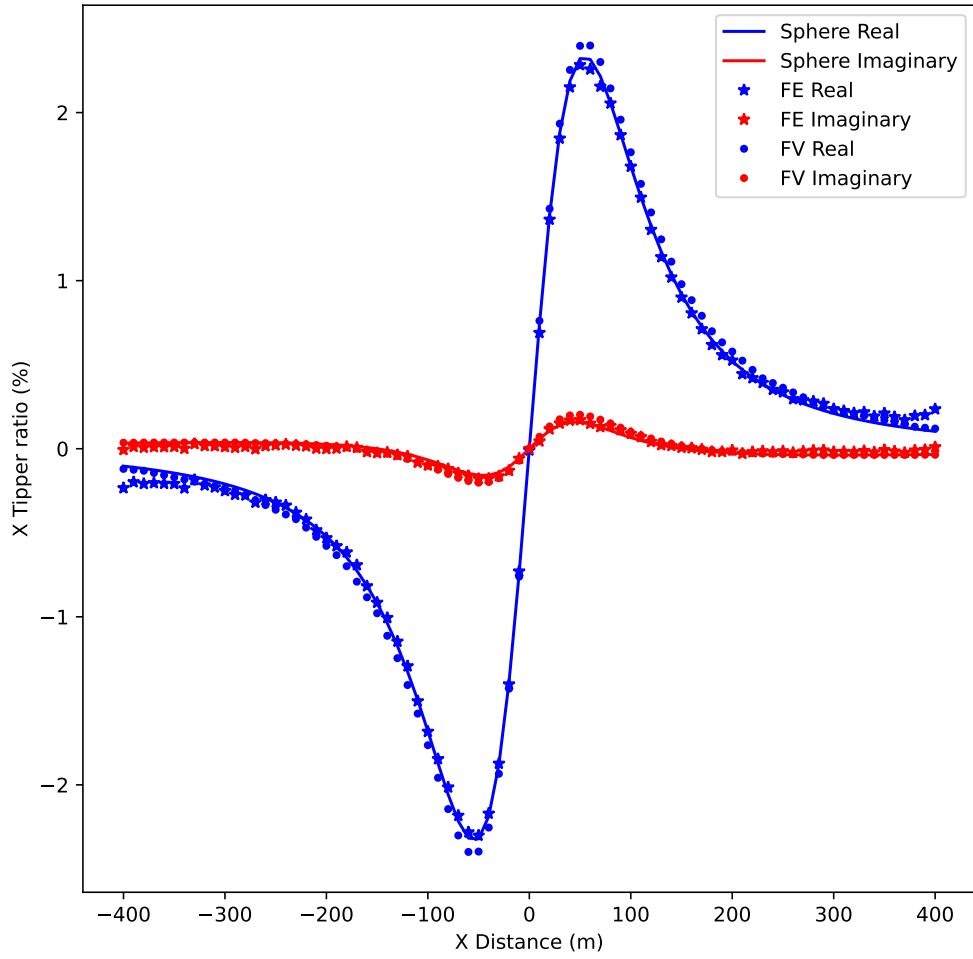


Figure 14: Comparison of the magnetic transfer function response (tipper) of a sphere buried in a layered-Earth model predicted with the sphere, FE, and FV models.

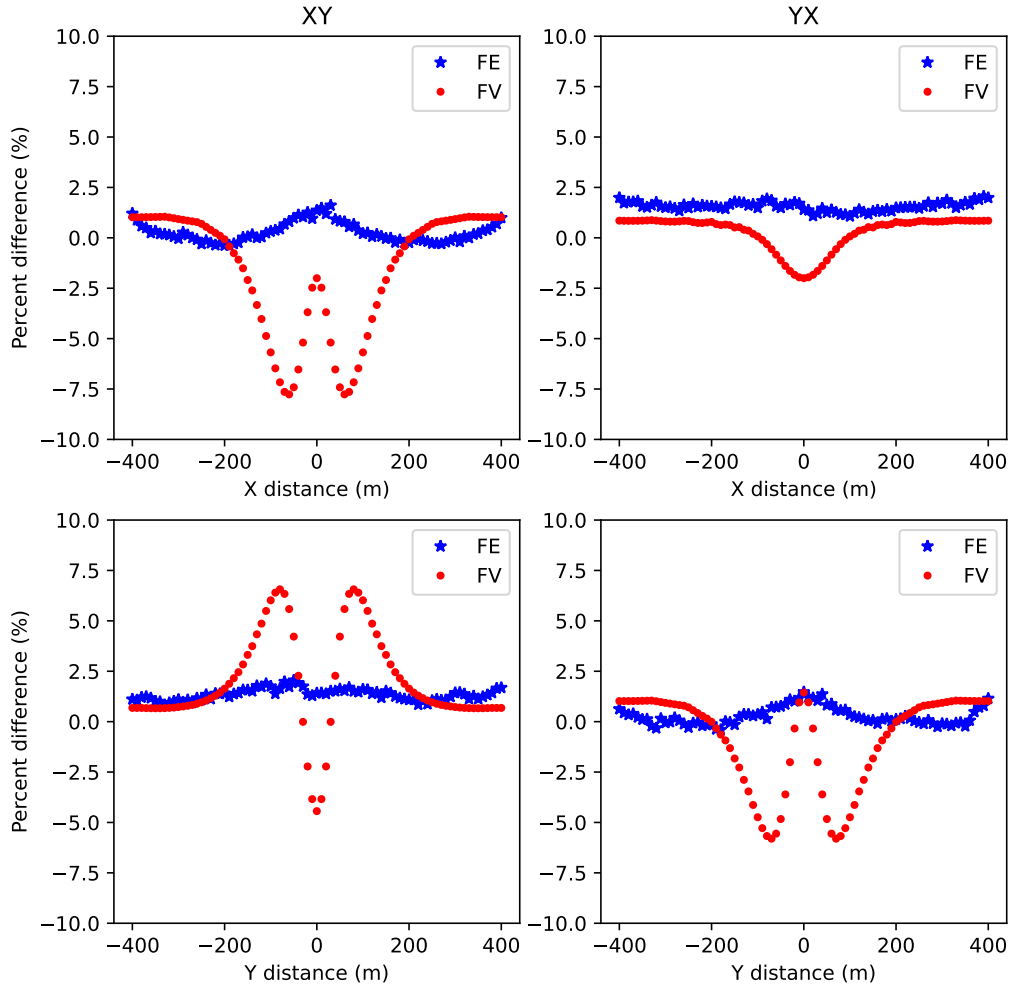


Figure 15: Percentage difference relative to the sphere model for the FE and FV approaches modelling a sphere in an homogeneous half-space.

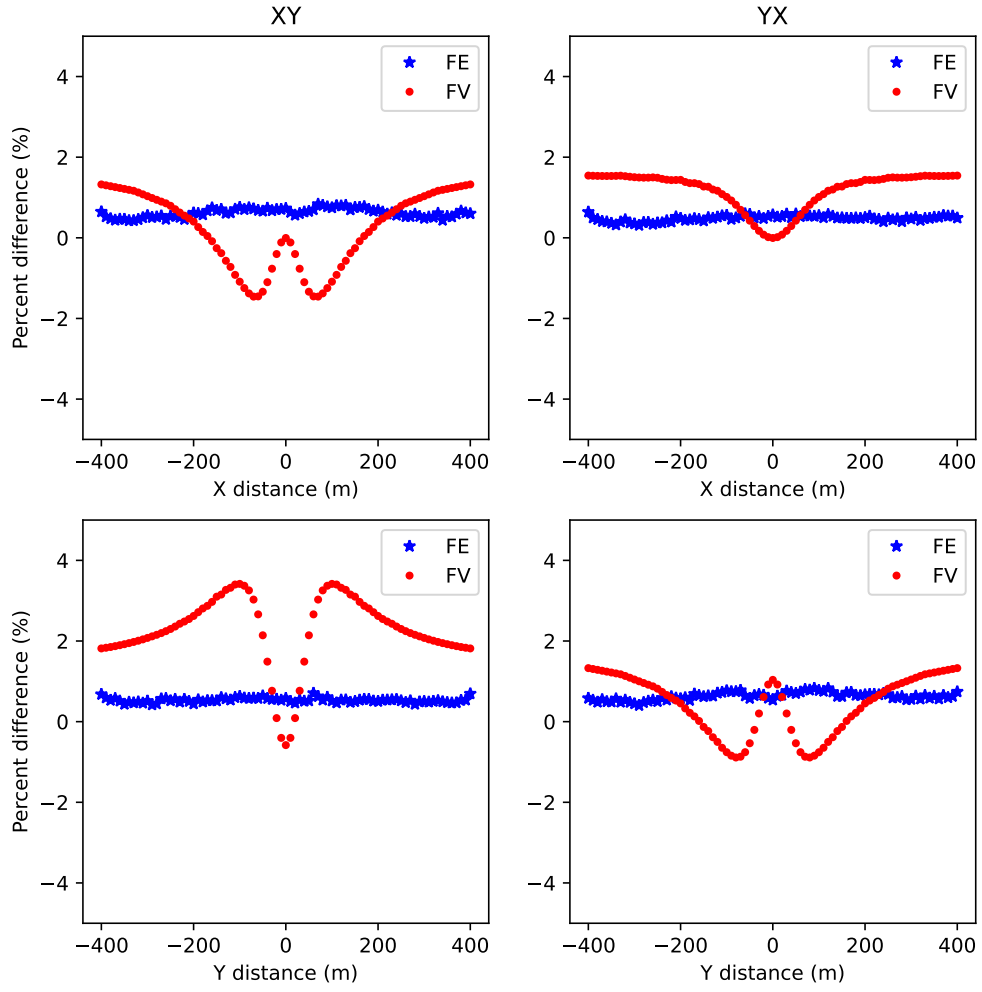


Figure 16: Percentage difference relative to the sphere model for the FE and FV approaches modelling a sphere in a layered earth.